Iterative SAT-Based Post-Processing for Collision-Free 3D Indoor Scene Generation

Anonymous Authors

Affiliation withheld for double-blind review

Abstract. Recent advancements in 3D Indoor Scene Generation have demonstrated impressive capabilities in synthesizing realistic indoor environments. However, scenes generated by diffusion models often contain physically implausible interpenetrations between furniture objects and walls. We present a lightweight iterative post-processing method that resolves these collisions without retraining the generative model. Leveraging DiffuScene as the baseline, our method applies a deterministic collision resolver algorithm to the predicted layout. We integrate an alignment-biased axis-selection strategy based on the Separating Axis Theorem together with boundary-aware constraints to resolve object collisions while keeping furniture within room boundaries. The solver iteratively alternates between boundary containment and oriented-box collision repair for at most 500 steps until it reaches convergence. Experimental results demonstrate significantly reduced object- and scene-level collision rates while maintaining comparable FID and KID scores. This highlights the potential of our collision resolver as a dedicated post-processing module for improving the physical plausibility of diffusion-based 3D scene generation methods.

Keywords: 3D Scene Generation, Collision Resolver, Separating Axis Theorem, Post-processing

1 Introduction

3D Indoor Scene Generation is a foundational task in computer graphics, virtual reality, and synthetic data generation for embodied AI. Since 2024, diffusion models have emerged as the leading paradigm for this task. State-of-the-art models like CommonScenes, [1] EchoScene, [2] and DiffuScene [3] leverage continuous diffusion processes to arrange 3D objects realistically based on learned spatial distributions, outputting a set of oriented bounding boxes (OBBs) that dictate the position, scale, and orientation of furniture objects in a 3D scene.

Challenges: Despite their visual appeal and strong semantic layout understanding, a persistent challenge with unconstrained 3D diffusion models is the lack of strict physical guarantees. Because the diffusion training objective typically minimizes a reconstruction loss over bounding box parameters, it does not explicitly penalize inter-object collisions or out-of-room placements. Existing methods that aim to address these inaccuracies typically involve retraining the network entirely with differentiable collision constraints. This approach is computationally expensive, requires access to model weights, and can destabilize the training process.

While recent work such as PhyScene [4] has attempted to improve physical plausibility by incorporating collision-aware objectives into diffusion model training or by applying gradient-based guidance during inference, these approaches introduce several limitations. First, modifying the training objective can interfere with the model's ability to learn the underlying semantic distribution of indoor layouts, potentially reducing generative diversity. Second, retraining or fine-tuning diffusion models is computationally expensive and often impractical, particularly for large-scale pretrained models, and any such modification is tied to a single architecture, offering no guarantee of transfer to other generative frameworks.

Our Approach: To address these limitations, we propose a *Collision Resolver*, a lightweight and fully training-free framework for enforcing physical plausibility in generated 3D scene layouts. Unlike approaches that incorporate physical constraints during training or the diffusion process, our method operates exclusively as a post-processing step on the final denoised layout, directly resolving inter-object collisions and invalid furniture placements. Operating entirely at inference time, our method requires no retraining and therefore preserves the generative model's learned distribution. This strict separation from the generation process makes our approach inherently model-agnostic and enables it to be readily applied to layouts produced by different 3D scene generation architectures.

We instantiate and validate this idea using DiffuScene [3] as our primary baseline, applying a deterministic, alignment-biased SAT-based Collision Resolver directly to its generated bounding boxes. Crucially, because the resolver operates purely on layout geometry - object extents, positions, and orientations - rather than on any pipeline-specific representation, the same module can be dropped into the output stage of other layout-generating scene synthesis methods with no retraining. We verify this by applying our resolver, without modification, to layouts produced by CommonScene [1] - a dual branch diffusion model with a distinct layout generation mechanism from DiffuScene - and observe consistent elimination of inter-object collisions and boundary violations across the two pipelines. This cross-pipeline consistency is central to our contribution: rather than proposing a collision-resolution heuristic tuned to one model's failure modes, we show that treating collision resolution as a purely geometric, post-hoc problem yields a solver that is reusable across architecturally distinct scene synthesis methods.

Concretely, this paper makes the following contributions:

- We formulate collision resolution for generated indoor scenes as a deterministic, training-free post-processing problem, decoupling physical plausibility entirely from the generative model's learning objective.
- We introduce an alignment-biased extension to the classical Separating Axis Theorem that jointly enforces room-boundary containment and resolves pairwise OBB collisions. This ultimately removes inter-object collisions and minimizes unnatural displacement of furniture.
- We demonstrate that this resolver generalizes across two 3D scene generation pipelines - CommonScenes, and DiffuScene - achieving consistent, substantial reductions in object- and scene-level collision rates without any retraining.

2 Related Work

Diffusion-Based Scene Generation. Recent advances in deep generative models have substantially improved the realism and diversity of 3D indoor scene synthesis [5]. Early approaches relied on autoregressive or transformer-based architectures, such as ATISS [6], while more recent methods formulate scene generation as a diffusion process over object layouts. Existing method like DiffuScene [3] generates indoor scenes directly from natural language descriptions by progressively denoising a set of oriented bounding boxes using cross-attention between language features and the diffusion model. Other conditional generation frameworks, including CommonScenes,[1] EchoScene, [2] and InstructScene [7], instead exploit structured scene graphs or relational representations to guide layout synthesis. Despite producing semantically coherent layouts, these methods primarily optimize semantic consistency rather than explicit physical feasibility.

Physical-Plausibility in 3D Layouts. Although modern diffusion models produce visually convincing indoor layouts, they frequently violate basic physical constraints, including object interpenetration and boundary violations. Recent efforts attempt to address these issues by incorporating physical objectives directly into the generative process. PhyScene, [4] for example, integrates collision-aware objectives into diffusion training and employs gradient-based guidance during inference to encourage physically valid layouts. While effective, such approaches require modifying the underlying generative model, which increases training complexity and computational cost. In contrast, our method completely decouples physical reasoning from semantic generation by treating collision resolution as a deterministic post-processing problem.

Collision Resolution. Traditional collision resolution methods formulate layout refinement as a continuous optimization problem, iteratively minimizing collision energy using gradient-based optimization [4]. Other approaches employ physics simulators such as PyBullet [8] MuJoCo [9] to resolve interpenetrations through rigid-body simulation. Although physically accurate, these approaches are computationally expensive for static indoor scene layouts. Our method instead adopts a lightweight Separating Axis Theorem (SAT) approach, providing an efficient geometric formulation that directly computes collision-free translations without iterative gradient optimization or full physics simulation.

3 Methodology

Given a scene layout generated by a 3D scene generation baseline (e.g., DiffuScene [3]), represented as a set of oriented bounding boxes, our goal is to produce a collision-free 3D scene that satisfies fundamental physical constraints.

To this end, we introduce a fine-grained collision resolution pipeline that eliminates object interpenetrations through alternating room-bound containment and Separating Axis Theorem (SAT)-based collision repair, as illustrated in Figure 1. Given a text prompt describing the desired scene, the baseline generation model first synthesizes an initial layout of oriented bounding boxes and retrieves or generates the corresponding 3D object meshes. These 3D object meshes are positioned, scaled, and rotated according to the predicted layout. Our *Collision Resolver* then operates purely on this set of oriented

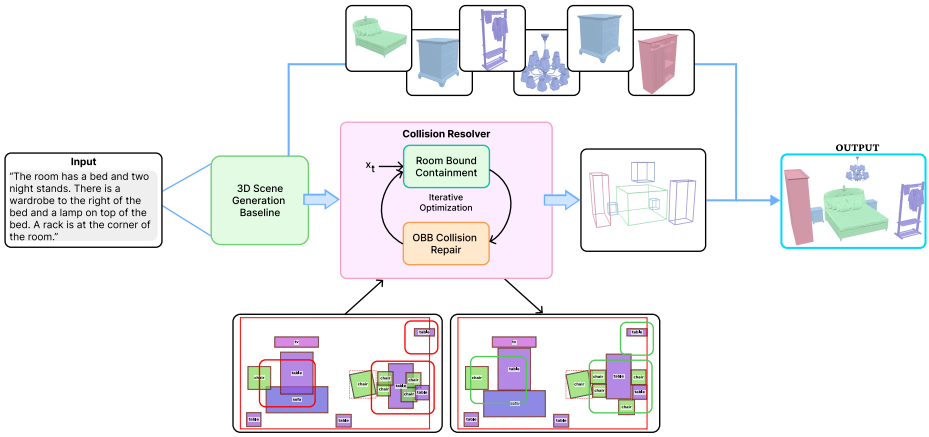


Fig. 1. Overview of the proposed Collision Resolver. Given a user-provided text prompt, a 3D scene generation baseline (eg., DiffuScene)will first generates a scene layout represented by oriented bounding boxes and retrieves the corresponding 3D object meshes. Our collision resolver then iteratively alternates between roombound containment and OBB collision repair using the Separating Axis Theorem (SAT) until convergence, i.e., no collisions remain in the scene. The refined bounding boxes are finally used to produce a physically plausible, collision-free 3D scene while preserving the original semantic layout. The bottom-left panel shows an input layout generated by DiffuScene, where the red boxes highlight object interpenetrations and out-of-bound placements. The bottom-right panel shows the corresponding layout after applying our SAT-based collision resolver, with the resolved regions highlighted in green.

bounding boxes, iteratively alternating between room-bound containment and SAT-based collision repair until convergence, i.e., until no collisions between objects remain. The refined bounding-boxes are finally reapplied to the retrieved meshes, yielding a physically plausible, collision-free 3D scene that preserves the semantic layout intended by the baseline model.

3.1 Preliminaries

DiffuScene [3] formulates indoor scene synthesis as a conditional denoising diffusion process over scene layouts. Starting from Gaussian noise, the diffusion model progressively predicts a clean set of object parameters conditioned on the room type and scene context. At the end of the reverse diffusion process, the model produces a structured scene layout consisting of M oriented bounding boxes, each corresponding to a furniture instance.

Each bounding box is parameterized as

$$\mathbf{b}_i = (l_i, h_i, w_i, x_i, y_i, z_i, \sin\theta_i, \cos\theta_i) \in \mathbb{R}^8, \tag{1}$$

where (x_i, y_i, z_i) denotes the center position $\mathbf{p}_i$, (l_i, h_i, w_i) denotes the object dimensions, and θ_i is the yaw orientation. We further define an *objectness mask* $\mathbf{o} \in \{0, 1\}^M$ to distinguish valid furniture instances ($o_i = 1$) from auxiliary nodes such as the floor and room boundary.

3.2 Post-Processing Collision Resolver

Because diffusion-based layout generators such as DiffuScene do not enforce any implicit or explicit physical constraints during the denoising process, the resulting bounding-box layouts frequently exhibit inter-object penetrations and out-of-bound placements. We address this with our *Collision Resolver* - a deterministic post-processing stage (Algorithm 1) that iteratively removes inter-object collisions and invalid object placements, producing a physically-plausible 3D scene directly at inference time.

The resolver first performs a fast 1D vertical bounding-box check along the y-axis: object pairs that are vertically separated (e.g., a monitor resting on a desk) require no further correction and are immediately excluded. For every remaining objects, vertically overlapping pair, resolution proceeds entirely in the 2D floor plane (xz), combining (a) *oriented room-bound containment clamping*, and (b) *OBB-SAT collision detection and resolution*, which generalizes the classical Minimum Translation Vector (MTV) through an alignment-biased axis selection.

Oriented Room-Bound Containment Object placements produced by 3D diffusion models are not guaranteed to lie within the room footprint, and a naive axis-aligned clamp is incorrect whenever the room itself is rotated with respect to the world frame. We therefore constrain every furniture instance within the room boundary expressed in the room's *own* oriented frame, rather than in world coordinates.

We first isolate the auxiliary layout nodes $\mathcal{L} = \{i : o_i = 0 \ \vee \ \ell_i = 0\}$, where $\ell_i = \arg\max_c \mathbf{y}_i^{(c)}$ is the predicted class index of box i and class 0 denotes floor/room instances. When a scene contains several such instances, we take the one with the largest footprint area,

$$L^* = \arg\max_{i \in \mathcal{L}} \ l_i \, w_i, \tag{2}$$

and read off its center $\mathbf{c}_L$, half-extents $(h_{xL}, h_{zL}) = (l_{L^*}/2, w_{L^*}/2)$, and oriented axes $\hat{\mathbf{a}}_L = (\cos\theta_{L^*}, \sin\theta_{L^*})$, $\hat{\mathbf{a}}_L^{\perp} = (-\sin\theta_{L^*}, \cos\theta_{L^*})$. Alternatively, for models such as DiffuScene that are conditioned on a fixed room geometry, we directly compute the bounding box of the ground-truth floor plan vertices, yielding an axis-aligned room frame with center $\mathbf{c}_L = (0, 0)$, half-extents (h_{xL}, h_{zL}), and identity axes $\hat{\mathbf{a}}_L = (1, 0), \hat{\mathbf{a}}_L^{\perp} = (0, 1)$. Together, these define the room's local frame. If neither a layout node nor a floor plan is available, the containment step is skipped for that scene.

For every non-layout object $i \notin \mathcal{L}$, we express its four floor-plane corners C_i in this local frame,

$$u = (C_i - \mathbf{c}_L) \cdot \hat{\mathbf{a}}_L, \qquad v = (C_i - \mathbf{c}_L) \cdot \hat{\mathbf{a}}_L^{\perp}, \tag{3}$$

and compute a per-axis correction $(\Delta u, \Delta v)$ that pulls any out-of-bound corners back within the half-extents. Along u,

$$\Delta u = \begin{cases} -\dfrac{u_{\max} + u_{\min}}{2}, & u_{\max} - u_{\min} > 2h_{xL}, \\ h_{xL} - u_{\max}, & u_{\max} > h_{xL}, \\ -h_{xL} - u_{\min}, & u_{\min} < -h_{xL}, \\ 0, & \text{otherwise}, \end{cases} \tag{4}$$

Algorithm 1 Post-Processing SAT Collision Resolver

Require: Layout boxes $\{\mathbf{b}_i\}_{i=1}^{M}$ with centers $\mathbf{p}_i$; mask $\mathbf{o}$; class indices $\{\ell_i\}$; max iterations $K_{\max} = 500$; clearance $\varepsilon_{\text{push}}$; bias β
Ensure: Collision-free layout $\{\mathbf{b}'_i\}$

$\mathcal{L} \leftarrow \{i : o_i = 0 \ \vee \ \ell_i = 0\}$ {auxiliary floor/layout nodes}
$L^* \leftarrow \arg\max_{i \in \mathcal{L}} \ l_i \, w_i$ {largest-footprint reference room box}
for $k = 1, 2, \ldots, K_{\max}$ **do**
  converged $\leftarrow$ **true**
  {1. Oriented room-bound containment (Section 3.2)}
  if $\mathcal{L} \neq \varnothing$ **then**
    for each object $i \notin \mathcal{L}$ **do**
      Project corners C_i into the room frame of L^* via Eq. (3)
      Compute correction $(\Delta u, \Delta v)$ via Eq. (4)
      if $(\Delta u, \Delta v) \neq (0, 0)$ **then**
        Update $\mathbf{p}_i$ via Eq. (5); *converged* $\leftarrow$ **false**
      end if
    end for
  end if
  {2. Object-object collision resolution}
  for each pair (i, j) with $i < j$, $o_i = o_j = 1$ **do**
    if vertically separated (1D y-axis check) **then**
      continue
    end if
    Compute corner sets C_i, C_j in xz-plane
    Compute overlap depths $\{\delta_k\}_{k=1}^{4}$ via Eq. (6)
    if $\exists\, k : \delta_k \leq 0$ **then**
      continue {no collision}
    end if
    $\hat{\mathbf{c}}_{ij} \leftarrow (\mathbf{p}_j - \mathbf{p}_i)/\|\mathbf{p}_j - \mathbf{p}_i\|$
    Select push axis $\hat{\mathbf{a}}^*$ and depth δ^* via Eq. (7)
    $\Delta \leftarrow (\delta^* + \varepsilon_{\text{push}})/2$
    $\mathbf{p}_i \leftarrow \mathbf{p}_i - \Delta \cdot \hat{\mathbf{a}}^*$
    $\mathbf{p}_j \leftarrow \mathbf{p}_j + \Delta \cdot \hat{\mathbf{a}}^*$
    converged $\leftarrow$ **false**
  end for
  if *converged* **then**
    break
  end if
end for
return $\{\mathbf{b}'_i\}$

with Δv defined analogously along $\hat{\mathbf{a}}_L^{\perp}$ using h_{zL}. The first branch handles objects whose footprint exceeds the room extent along that axis (e.g. an oversized asset), in which case the object is re-centered on the room rather than driven against a single wall; the remaining branches clamp the object to whichever wall it protrudes through, and leave objects that are already contained unchanged. The world-space center is then updated as

$$\mathbf{p}_i \leftarrow \mathbf{p}_i + \Delta u \, \hat{\mathbf{a}}_L + \Delta v \, \hat{\mathbf{a}}_L^{\perp}. \tag{5}$$

This step is applied to all non-layout objects at the start of every outer iteration k (Algorithm 1), *before* pairwise SAT collision resolution. As a result, the boundary-penalized overlap depth used by the alignment-biased axis selection (Section 3.2, Eq. (7)) is always evaluated against an up-to-date, already-contained room configuration, so that object-object pushes and room clamping will cooperate rather than fight each other across iterations.

OBB Collision Detection via SAT We detect collisions between oriented bounding boxes (OBBs) using the Separating Axis Theorem (SAT) [10]. For two OBBs with yaw angles θ_i, θ_j, we test four candidate separating axes.

Let $\mathbf{a}_k$ ($k = 1, \ldots, 4$) denote these axes, and let C_i, C_j denote the sets of four floor-plane corners of boxes i and j. For each axis $\mathbf{a}_k$, we project both sets of corners and compute the *overlap depth*:

$$\begin{aligned}\delta_k = \min\big(&\max(C_i \cdot \mathbf{a}_k), \max(C_j \cdot \mathbf{a}_k)\big) \\ &- \max\big(\min(C_i \cdot \mathbf{a}_k), \min(C_j \cdot \mathbf{a}_k)\big).\end{aligned} \tag{6}$$

If $\delta_k \leq 0$ for any axis, the boxes are not colliding. Otherwise, all four axes yield positive overlaps.

Alignment-Biased Axis Selection The standard SAT approach selects the axis with the minimum overlap depth $\delta^* = \min_k \delta_k$. For interior layouts, this often produces unintuitive sliding. We introduce an *alignment-biased* axis selection that prefers axes aligned with the center-to-center direction $\mathbf{c}_{ij} = \mathbf{p}_j - \mathbf{p}_i$.

Each axis $\mathbf{a}_k$ is oriented to point from i toward j: $\hat{\mathbf{a}}_k = \text{sign}(\mathbf{a}_k \cdot \mathbf{c}_{ij}) \cdot \mathbf{a}_k$. We compute an alignment score:

$$\alpha_k = \left|\hat{\mathbf{a}}_k \cdot \hat{\mathbf{c}}_{ij}\right|, \qquad s_k = \frac{\tilde{\delta}_k}{\alpha_k^{\beta} + \epsilon} \tag{7}$$

where $\beta = 2$ and ϵ is a small constant. Crucially, $\tilde{\delta}_k$ represents a *boundary-penalized* overlap depth, computed with respect to the same reference room frame L^* used in Section 3.2. For an object with corners C, define its total room-boundary violation as

$$g(C) = \max(0, u_{\max} - h_{xL}) + \max(0, -h_{xL} - u_{\min}) + \max(0, v_{\max} - h_{zL}) + \max(0, -h_{zL} - v_{\min}), \tag{8}$$

using the projections (u, v) of Eq. (3). Before penalizing an axis, we tentatively apply the half-depth push $\pm(\delta_k/2)\hat{\mathbf{a}}_k$ to the corner sets C_i, C_j and compare $g(\cdot)$ before and after: if either object's violation *increases* ($g_{\text{after}} - g_{\text{before}} > 10^{-3}$), the axis is deemed to push an object further out of the room, and we set $\tilde{\delta}_k = 1000\,\delta_k$; otherwise $\tilde{\delta}_k = \delta_k$. This preemptive check prevents the object-object resolution step from undoing the room containment of Section 3.2 at the next iteration. Axes with $\alpha_k < 0.15$ are discarded. The axis with the lowest score s_k is selected as the push direction.

Given the selected push axis $\hat{\mathbf{a}}^*$ and depth δ^* for a colliding pair (i, j), we symmetrically displace both objects by $(\delta^* + \varepsilon_{\text{push}})/2$ along $\hat{\mathbf{a}}^*$:

$$\mathbf{p}_i \leftarrow \mathbf{p}_i - \frac{\delta^* + \varepsilon_{\text{push}}}{2}\,\hat{\mathbf{a}}^*, \qquad \mathbf{p}_j \leftarrow \mathbf{p}_j + \frac{\delta^* + \varepsilon_{\text{push}}}{2}\,\hat{\mathbf{a}}^* \tag{9}$$

where $\varepsilon_{\text{push}} = 0.02\,\text{m}$ provides clearance. Algorithm 1 iterates over all pairs until no collisions remain or $K_{\max} = 500$ is reached.

4 Experiments and Results

To evaluate the effectiveness of our SAT-based collision resolver, we apply it to the outputs of two representative layout generators (CommonScene [1], and DiffuScene [3]). The evaluation focuses on two complementary axes: physical plausibility and preservation of generative visual quality.

4.1 Experimental Setup

Dataset. We conduct all experiments on the 3D-FRONT dataset [11], following the evaluation protocol of CommonScene [1] and DiffuScene [3]. We evaluate three common indoor room categories: Bedroom, LivingRoom, and DiningRoom. Collision metrics are computed on 162 bedroom, 192 livingroom and 177 diningroom test scenes, respectively.
Evaluation Metrics. We evaluate physical plausibility using **ColObj** and **ColScene**, the two collision metrics introduced by PhyScene [4]. Importantly, collisions are evaluated on the final *retrieved 3D meshes*, rather than on the oriented bounding boxes (OBBs) used internally by our collision resolver. For each scene, we first use axis-aligned bounding-box (AABB) overlap as a pre-filter for candidate object pairs. Each candidate pair is then evaluated at the mesh level using a point-in-mesh occupancy test (Kaolin's inside/outside check [12] when available, with a ray-casting fallback). Two objects are considered colliding if any vertex of one mesh lies inside the other.

Let M denote the number of objects in a scene and $c_i \in \{0, 1\}$ indicate whether object i is involved in at least one mesh-level collision. We report

$$\text{ColObj} = \frac{1}{M}\sum_{i=1}^{M} c_i, \qquad \text{ColScene} = \mathbb{1}\left[\sum_{i=1}^{M} c_i > 0\right]. \tag{10}$$

ColObj measures the fraction of furniture instances involved in at least one collision, while **ColScene** indicates whether a scene contains any colliding pair; both are averaged over all scenes in each room category.

To assess whether collision resolution preserves generative quality, we additionally report Fréchet Inception Distance (FID) [13] and Kernel Inception Distance (KID) [14], following the standard of prior 3D scene generation methods.

4.2 Quantitative Results

Physical Plausibility Table 1 reports ColObj and ColScene for CommonScenes, and DiffuScene, both with their raw generated/retrieved layouts and after applying our *Collision Resolver*. Because the resolver iterates until no pairwise SAT overlap remains, ColObj and ColScene are driven to near zero.

Convergence and Stability. Table 2 and Figure 2 summarize the convergence behavior and displacement cost of our solver. The resolver efficiently eliminates violations

Table 1. Physical Plausibility: Collision and Out-of-Boundary Metrics (↓).

Method	Bedroom			Living Room			Dining Room		
	ColObj$_{\downarrow}$	ColScene$_{\downarrow}$	OOB$_{\downarrow}$	ColObj$_{\downarrow}$	ColScene$_{\downarrow}$	OOB$_{\downarrow}$	ColObj$_{\downarrow}$	ColScene$_{\downarrow}$	OOB$_{\downarrow}$
ATISS [6]	0.255	0.500	–	0.372	0.870	–	0.483	0.910	–
PhyScene [4]	0.187	0.350	–	0.130	0.477	–	0.134	0.400	–
CommonScene [1]	0.352	0.628	–	0.312	0.673	–	0.386	0.810	–
CommonScene + Resolver (Ours)	**0.021**	**0.043**	–	**0.009**	**0.032**	–	**0.137**	**0.043**	–
DiffuScene [3]	0.267	0.469	2.191	0.302	0.797	4.703	0.314	0.751	4.542
DiffuScene + Resolver (Ours)	**0.000**	**0.000**	**0.444**	**0.61**	**3.12**	**0.823**	**1.42**	**7.34**	**1.362**

†OOB reported only for DiffuScene due to time constraints; full evaluation of CommonScenes left to future work.

Table 2. Convergence and displacement cost of the SAT resolver.

Room	Colliding Pairs	Mean Depth (cm / % obj.)	Mean Iters.	Objects Moved	Mean Disp. (cm / % obj.)
Bedroom	15	23.9 ± 25.7 / 51.5%	1.7 ± 0.7	67.1%	33.7 ± 28.7 / 77.6%
Living Room	65	20.8 ± 21.5 / 52.4%	6.5 ± 36.0	75.4%	73.7 ± 72.3 / 169.7%
Dining Room	70	22.0 ± 21.0 / 53.8%	18.5 ± 82.3	80.1%	81.1 ± 80.2 / 201.5%

for the majority of scenes: 100% of bedroom scenes converge within 5 iterations, while over 75% of living room and dining room scenes converge in the same budget. A small fraction of highly congested layouts requires more iterations, with only 2.8% of dining rooms and 0.5% of living rooms hitting the maximum budget of $K_{max} = 500$. The displacement metrics indicate that $67 - 80\%$ of objects are adjusted by $33 - 81$ cm on average. This relatively high displacement primarily stems from the room-bound containment step, which strictly enforces that objects generated outside the valid floor plan are pulled back into the room, ensuring physical validity.

Robustness under Adversarial Density. To probe robustness beyond the typical dataset operating regime, we stress-test the resolver on 50 synthetic layouts with artificially elevated object density (20 objects per $5\text{m} \times 5\text{m}$ room, roughly double the typical furniture count), yielding an average of 48.3 collisions per scene prior to resolution. Even under this adversarial density, the resolver successfully converges for all scenes (0/50 failures), requiring an average of 14.1 iterations (maximum 19). This confirms the stability of the alternating resolution strategy under challenging, highly congested conditions.

Visual Fidelity It is critical that correcting physical layouts does not destroy the learned semantic distribution of the generative model. Table 3 reports FID and KID for DiffuScene before and after our post-processing.[1]

Both FID and KID increase slightly after post-processing, more noticeably in the Dining Room category (KID: 0.72 → 2.16) than in the Bedroom category (KID:

[1] FID/KID for ATISS+SAT and PhyScene+SAT were not completed at submission time due to the additional rendering cost of regenerating image sets for each baseline; we report generative-quality preservation only for the DiffuScene baseline and leave a full cross-method comparison to future work.

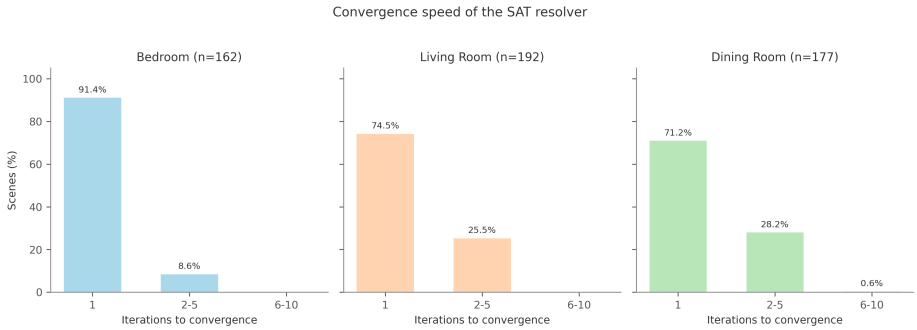


Fig. 2. Distribution of iterations required for the SAT resolver to converge, per room category. Most scenes converge within 5 iterations, with a long tail for highly congested layouts.

Table 3. Generative Quality: FID and KID (↓)

Method	**Bedroom**		**Dining Room**	
	FID↓	**KID↓**	**FID↓**	**KID↓**
DepthGAN [15]	40.15	18.54	81.13	50.63
Sync2Gen [16]	31.07	11.21	46.05	8.74
ATISS [6]	18.60	1.72	38.66	5.62
DiffuScene (Baseline) [3]	**17.21**	**0.70**	32.60	**0.72**
Ours (DiffuScene + SAT)	18.79	0.96	**32.10**	2.16

0.70 → 0.96). We attribute this to a shift in the *relative proximity* between objects rather than a degradation of the overall layout: DiffuScene is trained to reproduce ground-truth arrangements in which furniture is frequently placed in close contact (e.g., dining chairs tucked under a table), and without an explicit non-penetration constraint, the model occasionally represents this closeness as slight OBB overlap. When our solver separates such pairs, it necessarily increases their inter-object distance by at least $\varepsilon_{\text{push}}$ (Eq. (9)), which can push the local spatial statistics of densely packed room types slightly outside the distribution the FID/KID Inception features were trained to recognize as "ground-truth-like". Dining rooms, which contain the highest density of small, closely spaced objects (chairs around a table) among our three categories, are the most affected by this effect, consistent with the trend in Table 3. Our alignment-biased axis selection (Section 3.2) is designed to minimize this shift by preferring pushes along the center-to-center direction rather than an arbitrary minimum-overlap axis, but it does not eliminate it entirely. We report FID/KID as a sanity check that the macro-level layout is not destroyed, rather than as evidence that the object-level spatial distribution is perfectly preserved; the latter would require a measurement more sensitive to local displacement, which we leave to future work (e.g., mean per-object displacement, reported per room category, and a paired significance test across multiple sampling seeds).

4.3 Qualitative Results

Figure 4 provides a representative visual assessment of the LivingRoom-88425 scene. The input layout contains several inter-object collisions and a boundary violation, highlighted by the red callouts. After post-processing, the solver separates the affected objects and restores room-bound containment, as indicated by the corresponding green callouts, while leaving the overall scene arrangement largely unchanged.

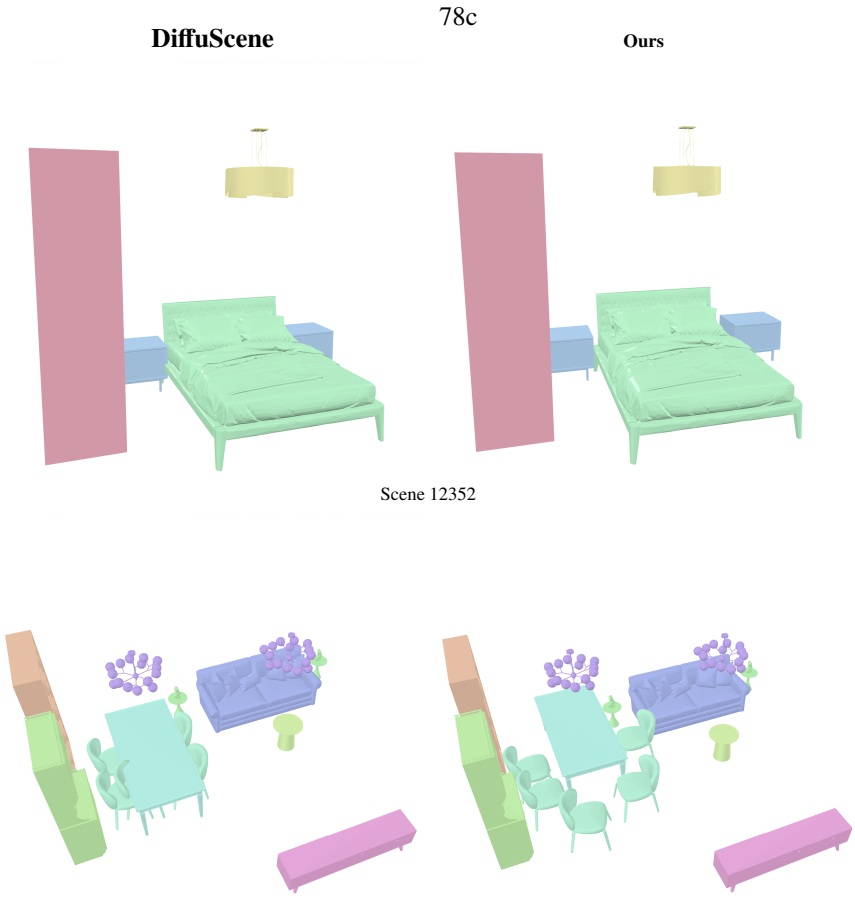
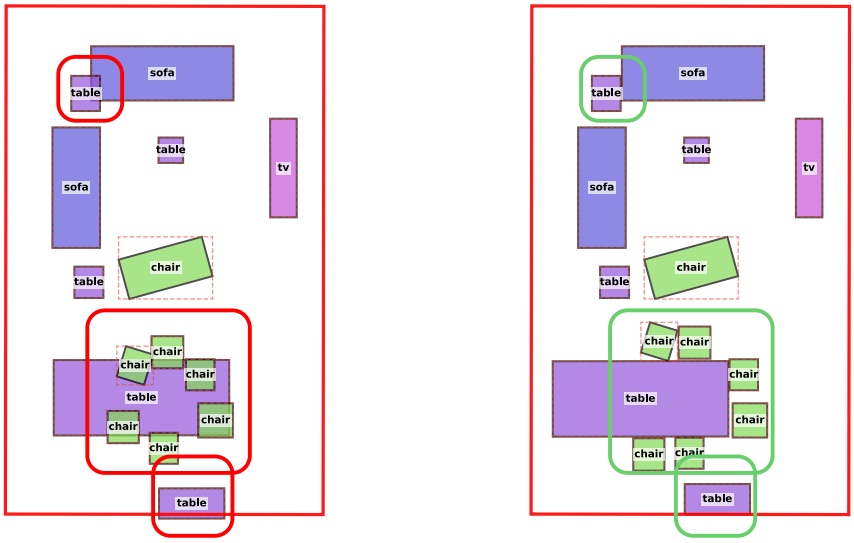


Fig. 3. Additional qualitative comparisons. The left column shows the original DiffuScene outputs, and the right column shows our collision-resolved results.

Figure 3 shows three additional before-and-after examples. The left image of each pair is the original DiffuScene output, while the right image is the corresponding result after applying our SAT-based collision resolver. The corrections remove visible interpenetrations while preserving the overall scene composition.



(a) Input layout **(b)** Resolved layout

Fig. 4. Visual comparison of SAT-based collision resolution. Red boxes indicate collisions and out-of-bound objects in the DiffuScene output, while green boxes show the corrected layout after SAT post-processing.

5 Conclusion

We introduced an efficient, training-free, post-processing methodology to resolve object collisions in 3D scenes generated by diffusion models. By utilizing an alignment-biased SAT solver bounded to 500 iterations, we effectively separate penetrating objects while maintaining room confines. Our experiments over the DiffuScene [3] baseline demonstrate a drastic reduction in collision rates while fully preserving the underlying aesthetic and generative quality, providing a practical solution for physically grounded scene generation.

Discussion and Limitations. The primary advantage of our SAT-based post-processing is its efficiency and model-agnostic nature. However, because it operates independently of the generative model's learned semantic priors, there is a minor trade-off. Pushing an object to resolve a collision might slightly shift it away from its semantically optimal position (e.g., moving a chair slightly too far from a dining table). To mitigate this, our alignment-biased axis selection ensures that the displacement is strictly minimized. As reflected in our unchanged FID and KID scores, these micro-adjustments successfully restore physical plausibility without visibly degrading the semantic coherence of the scene.

References

1. G. Zhai, E. P. Örnek, S.-C. Wu, Y. Di, F. Tombari, N. Navab, and B. Busam, "Commonscenes: Generating commonsense 3d indoor scenes with scene graph diffusion," *Advances in Neural Information Processing Systems*, vol. 36, pp. 30 026–30 038, 2023.
2. G. Zhai, E. P. Örnek, D. Z. Chen, R. Liao, Y. Di, N. Navab, F. Tombari, and B. Busam, "Echoscene: Indoor scene generation via information echo over scene graph diffusion," in *European Conference on Computer Vision*. Springer, 2024, pp. 167–184.
3. J. Tang, Y. Nie, L. Markhasin, A. Dai, J. Thies, and M. Nießner, "Diffuscene: Denoising diffusion models for generative indoor scene synthesis," in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024, pp. 20 507–20 518.
4. Y. Yang, B. Jia, P. Zhi, and S. Huang, "Physcene: Physically interactable 3d scene synthesis for embodied ai," in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024, pp. 16 262–16 272.
5. H. Xie, B. Wen, Z. Chen, F. Hong, and Z. Liu, "3d scene generation: A survey," *arXiv preprint arXiv:2505.05474*, 2025.
6. D. Paschalidou, A. Kar, M. Shugrina, K. Kreis, A. Geiger, and S. Fidler, "Atiss: Autoregressive transformers for indoor scene synthesis," *Advances in neural information processing systems*, vol. 34, pp. 12 013–12 026, 2021.
7. C. Lin and Y. Mu, "Instructscene: Instruction-driven 3d indoor scene synthesis with semantic graph prior," in *International Conference on Learning Representations*, vol. 2024, 2024, pp. 25 687–25 718.
8. E. Coumans and Y. Bai, "Pybullet, a python module for physics simulation for games, robotics and machine learning," 2016.
9. E. Todorov, T. Erez, and Y. T. MuJoCo, "A physics engine for model-based control," in *Proceedings of the 2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pp. 5026–5033.
10. S. Gottschalk, M. C. Lin, and D. Manocha, "Obbtree: A hierarchical structure for rapid interference detection," in *Proceedings of the 23rd annual conference on Computer graphics and interactive techniques*, 1996, pp. 171–180.
11. H. Fu, R. Jia, L. Gao, M. Gong, B. Zhao, S. Maybank, and D. Tao, "3d-future: 3d furniture shape with texture," *International Journal of Computer Vision*, vol. 129, no. 12, pp. 3313–3337, 2021.
12. C. F. Tsang, M. Shugrina, J.-F. Lafleche, O. Perel, C. Loop, T. Takikawa, V. Modi, A. Zook, J. Wang, W. Chen, T. Shen, J. Gao, K. M. Jatavallabhula, E. Smith, A. Rozantsev, S. Fidler, G. State, J. Gorski, T. Xiang, J. Li, M. Li, and R. Lebaredian, "Kaolin: A PyTorch Library for Accelerating 3D Deep Learning Research." [Online]. Available: https://github.com/NVIDIAGameWorks/kaolin
13. M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "Gans trained by a two time-scale update rule converge to a local nash equilibrium," *Advances in neural information processing systems*, vol. 30, 2017.
14. M. Bińkowski, D. J. Sutherland, M. Arbel, and A. Gretton, "Demystifying mmd gans," *arXiv preprint arXiv:1801.01401*, 2018.
15. Y. Li, Y. Wang, Z. Lu, and J. Xiao, "Depthgan: Gan-based depth generation of indoor scenes from semantic layouts," *arXiv preprint arXiv:2203.11453*, 2022.
16. H. Yang, Z. Zhang, S. Yan, H. Huang, C. Ma, Y. Zheng, C. Bajaj, and Q. Huang, "Scene synthesis via uncertainty-driven attribute synchronization," in *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2021, pp. 5610–5620.